Route Decay Model Card

HackFusion 2026 • M7 Predictive Route Decay • One-page submission export

Model Logistic regression • 2h impassability prediction • On-device JS inference in Expo

Metrics Precision 1.0 • Recall 0.8333 • F1 0.9091 • Threshold 0.7

Methodology

1. Sensor feed
1 Hz rainfall feed + edge context

→

2. Features
cumulative rain, rate change, elevation, soil pr

→

3. Training
80/20 split + standardization + logistic regress

→

4. Runtime
JSON coefficients m

Dataset + Features

Sensor feed: ml/training/rainfall_sensor_feed.csv
Training table: ml/training/route_decay_training.csv
Scenario graph: data/sylhet_map.json
Environment: data/edge_environment_context.json

Features:
- cumulative_rainfall_mm
- rainfall_rate_change
- elevation_m
- soil_saturation_proxy

Confusion Summary

TP: 5	FP: 0
TN: 44	FN: 1

Runtime integration

High-risk edges above 0.7 are penalized in the routing graph before route plans are served.
The backend and the Expo app both consume the same exported coefficient artifact family.